Practice Problems
Calculus Physics II

Name: Key **Date:** _____

Chapter 48 – Electromagnetic Induction

1) Electromagnetic induction occurs in a coil when there is a change in

A) the coil's polarity.
B) electric field intensity in the coil.
C) electromagnetic polarity.
(D) magnetic field intensity in the coil.
E) voltage in the coil.

$V_{ens} = -N \frac{\Delta \Phi}{\Delta t} = -N \frac{\Delta (\vec{B} A)}{\Delta t}$

2) Voltage can be induced in a wire by

A) moving a magnet near the wire. ✓
B) moving the wire near a magnet. ✓
C) changing the current in a nearby wire. ✓
(D) Choices A, B, and C are all true.
E) None of the above choices are true.

3) A changing magnetic field induces

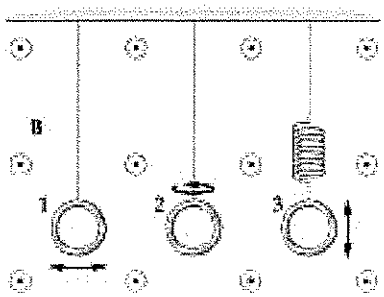
A) a magnetic field of greater magnitude.
B) a magnetic field of the same magnitude.
(C) an electric field.

4) When a bar magnet is thrust into a coil of copper wire, the coil tends to

(A) repel the magnet as it enters.
B) attract the magnet as it enters.
C) both of these
D) neither of these

5) The three loops of wire shown below are all subject to the same uniform magnetic field $\vec{B}$ that does not vary with time. Loop 1 oscillates back and forth as the bob in a pendulum, loop 2 rotates about a vertical axis, and loop 3 oscillates up and down at the end of a spring. Which loop, or loops, will have an induced emf?

A) Loops 2 and 3
B) Loops 1 and 3
C) Loop 1
(D) Loop 2
E) Loop 3



Loop # 2 is only coil that magnetic flux is changing.

6) A rectangular box has a length along the x axis of 0.5 m, a length along the y axis of 0.4 m and a length along the z axis of 0.3 m. If a uniform magnetic field of 2 T points along the positive z axis, what is the magnetic flux through the surface in the x-y plane?

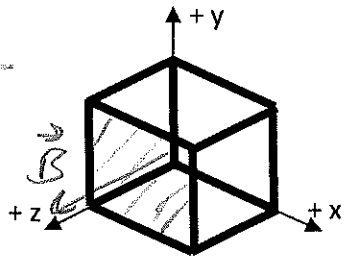
Area $= A = (x)(y) = (0.5\text{m})(0.4\text{m}) = 0.2\text{m}^2$

$\phi = 0^\circ$

Normal line to surface along $\vec{B}$-field

$\Phi = BA\cos\phi$

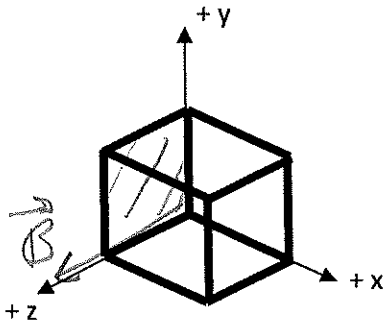
$\Phi = (0.4\text{ Wb})$



7) A rectangular box has a length along the x axis of 0.5 m, a length along the y axis of 0.4 m and a length along the z axis of 0.3 m. If a uniform magnetic field of 2 T points along the positive z axis, what is the magnetic flux through the surface in the y-z plane?

Normal line to $\vec{B}$-field is $\phi = 90^\circ$

$\Phi = BA\cos\phi = (0\text{ Wb})$

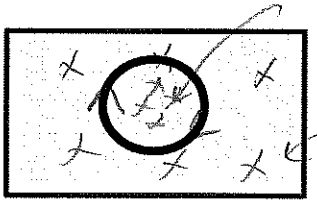


8) A coil lies flat on a horizontal tabletop in a region where the magnetic field points straight down. The magnetic field disappears suddenly. When viewed from above, what is the direction of the induced current in this coil as the field disappears?

B_{ext} is getting weaker

$\vec{B}_{ind}$ must add to $\vec{B}_{ext}$

A) counterclockwise
(B) clockwise
C) clockwise initially, then counterclockwise before stopping
D) There is no induced current in this coil.

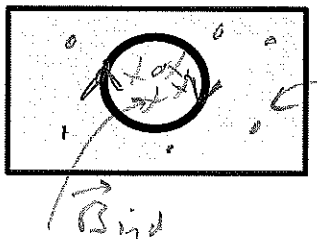


9) A coil lies flat on a horizontal tabletop a magnetic field suddenly appears and is pointing straight up out of the table. When viewed from above, what is the direction of the induced current in this coil as the field appears?

$\vec{B}_{ext}$ is getting stronger

$\vec{B}_{ind}$ must subtract from $\vec{B}_{ext}$

A) counterclockwise
(B) clockwise
C) clockwise initially, then counterclockwise before stopping
D) There is no induced current in this coil.

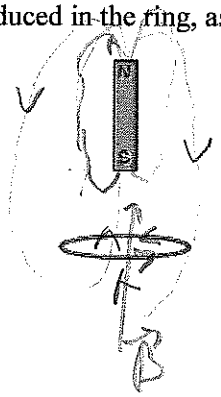


10) A bar magnet is oriented above a copper ring with its south pole pointing down, as shown in the figure. If the magnet is pulled upward, what is the direction of the current induced in the ring, as viewed from above?

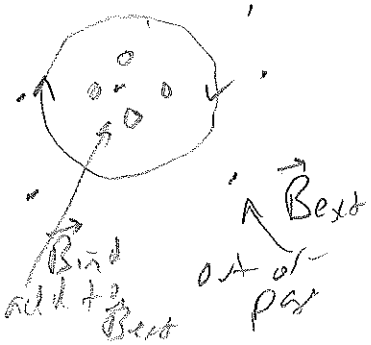
A) There is no current in the ring.
(B) counterclockwise
C) clockwise

$\vec{B}_{ext}$ points up page and is getting weaker.

$\vec{B}_{ind}$ must add to $\vec{B}_{ext}$



Top view

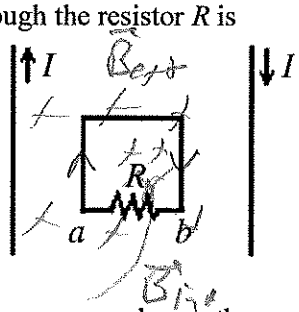


11) As shown in the figure, two parallel wires carry a current I in opposite directions, and this current is decreasing. A rectangular loop is centered between the wires. The direction of the induced current through the resistor R is

A) from a to b.
B) from b to a.
C) No current is induced.

Right hand rule # 2 From wire $\vec{B}_{ext}$ is into page and getting weaker.

$\vec{B}_{ind}$ must add to $\vec{B}_{ext}$



12) When a magnet is moved in and out of a wire coil, voltage is induced. If the coil has twice as many loops, the voltage induced is

A) half.
B) the same.
C) twice.
D) four times as much.
E) none of the above

$V_{ind} = -N\left(\frac{\Delta\Phi}{\Delta t}\right)$

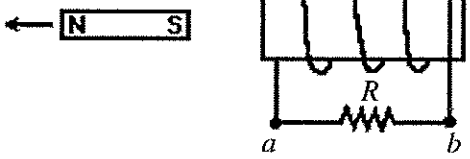
N = # of loops of wire

13) When a bar magnet is thrusted out of a coil of copper wire, the coil tends to

A) attract the magnet as it exits.
B) repel the magnet as it exits.
C) both of these
D) neither of these

14) In the figure, a bar magnet moves away from the solenoid. The direction of the induced current through the resistor R is

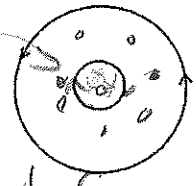
A) from b to a.
B) from a to b.
C) No current is induced.



15) An outer metal ring surrounds an inner metal ring, as shown in the figure. The current in the outer ring is counterclockwise and increasing. What is the direction of the induced current in the inner ring?

A) clockwise
B) counterclockwise
C) There is no induced current in the inner ring.

$\vec{B}_{ext}$ from outside loop out of page getting stronger



$\vec{B}_{ind}$ must subtract from $\vec{B}_{ext}$

16) The area of a rectangular loop of wire (N = 30 loops) is 2×10^{-3} m^2. The loop is placed in a magnetic field that changes from 0.20 T to 1.4 T in 0.02 s. The plane of the loop is perpendicular to the direction of the magnetic field. What is the magnitude of the induced emf in that loop?

$\Delta\Phi = B_f A_f \cos\phi_f - B_i A_i \cos\phi_i$

$\Delta\Phi = (1.4\,T)(2\times10^{-3}\,m^2)(1) - (0.2\,T)(2\times10^{-3}\,m^2)(1)$

$\Delta\Phi = 0.0024\ Wb$

$V_{ind} = -N\frac{\Delta\Phi}{\Delta t}$

$V_{ind} = -\frac{(30)(0.0024\,Wb)}{0.02\,s}$

$V_{ind} = -3.6\,V$

17) The area of a rectangular loop of wire (N = 10 loops) is 0.4 m^2, the loop surface is placed perpendicular to a magnetic field of 0.5 T. The normal to the loop is rotated from 90° to 30° in 0.03s. What is the magnitude of the induced emf in that loop?

$\Delta\Phi = B_f A_f \cos\phi_f - B_i A_i \cos\phi_i$

$\Delta\Phi = BA(\cos(\phi_f) - \cos(\phi_i))$

$\Delta\Phi = 0.173\ Wb$

$V_{emf} = -N\frac{\Delta\Phi}{\Delta t}$

$V_{emf} = -57.7\ V$

18) A coil of wire (N = 100 loops) surrounds an inner coil of wire (N = 100 loops), as shown in the figure. The outer coil has a radius of 0.1 m and no current flowing, the current is turned on and reaches a maximum of I = 20 A in a time of 2 x 10^{-4} s in the direction shown. What is the magnitude and direction of the induced current in the inner ring if it has a radius of r = 0.05 m and a resistance of 12 Ω? Assume the magnetic field generated by the outer ring is uniform throughout the inside of the ring.

$\vec{B}$ inside inner coil

$|\vec{B}| = \frac{\mu_0 I N}{2 r_{outer}} = 0.013T$

Area of inner coil

$A = \pi r^2 = 7.85 \times 10^{-3}\ m^2$

$\Delta\Phi = B_f A_f \cos(\phi_f) - B_i A_i \cos\phi_i$ ($\phi_i = 0$)

$V_{emf} = -N\frac{\Delta\Phi}{\Delta t} = -N\frac{(B_f A_f \cos\phi_f)}{\Delta t}$

$V_{emf} = 49.4\ V$

$V = IR$

$I = \frac{V_{emf}}{R} = \frac{49.4\ V}{12\ \Omega} = 4.1\ A$

19) A conducting loop (N = 1) is in the form of a circle and placed perpendicular to a magnetic field of 2.5 T. If the area of the loop increases from 0.3 m^2 to 0.8 m^2 in 0.1 s, what is the induced emf in the loop?

$\Delta\Phi = B_f A_f \cos\phi_f - B_i A_i \cos\phi_i$

$\Delta\Phi = B(A_f - A_i)(1) = 1.25\ Wb$

$V_{emf} = -N\frac{\Delta\Phi}{\Delta t}$

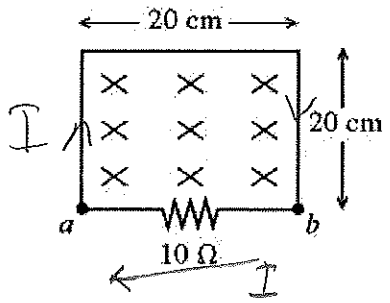
$V_{emf} = 12.5\ V$

20) As shown in the figure, a wire (N = 1) and a 10-Ω resistor are used to form a circuit in the shape of a square, 20 cm by 20 cm. A uniform magnetic field is directed into the plane of the circuit. The magnitude of the magnetic field is decreased from 2.50 T to 0.50 T in a time interval of 50 ms. Determine the induced current and its direction around the circuit.

$A = (0.2m)(0.2m) = 0.04m^2$

$\Delta\Phi = B_f A_f \cos\phi_f - B_i A_i \cos\phi_i$

$\Delta\Phi = A(B_f - B_i)(1)$



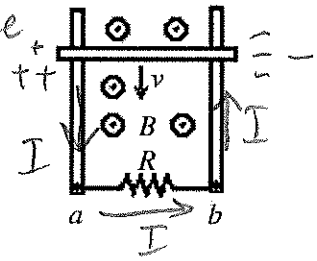
$V_{ind} = -N\frac{\Delta\Phi}{\Delta t} = -N\frac{(A(B_f - B_i))}{\Delta t} = 1.6V$

$V = IR \qquad I = \frac{V}{R} = \frac{1.6V}{10\Omega} = 0.16A$

21) As shown in the figure, a metal bar is in contact with a pair of metal parallel rails. A steady uniform magnetic field *B*, perpendicular to the plane of the rails and pointing outward from the page, is present. The bar is in downward motion with velocity of magnitude v. The direction of the induced current through the resistor *R* is

A) from *b* to *a*.
B) from *a* to *b*.
C) No current is induced.

Using right hand rule #1

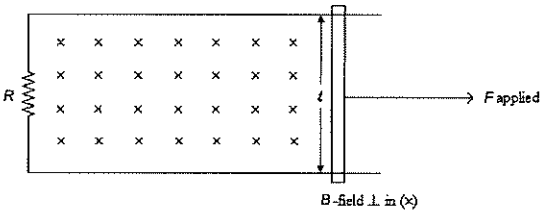


22) A conducting rod whose length ℓ = 25 cm is placed on a U-shaped metal wire that has a resistance *R* of 8 Ω as shown. The wire and the rod are in the plane of the paper. A constant magnetic field of strength 0.4 T is applied perpendicular and into the paper. An applied force moves the rod to the right with a constant speed of 6 m/s. What is the magnitude of the induced emf in the wire?

$L = 0.25m$
$|\vec{B}| = 0.4T$
$\phi = 90°$
$|\vec{v}| = 6m/s$

$V_{ind} = BLv\sin\phi$

$V_{ind} = 0.6V$



23) A car's axal is a conductor that can produce a induced voltage across it as the car moves in the earth's magnetic field. Assume the axal of the car is 2 m and the earth's magnetic field strength perpendicular to the axal is 3 x 10^{-5} T, what velocity must the car have in order to induce a potential difference of 1 V across the axal?

$|\vec{v}| = ?$

$L = 2m$

$|\vec{B}| = 3 \times 10^{-5} T$

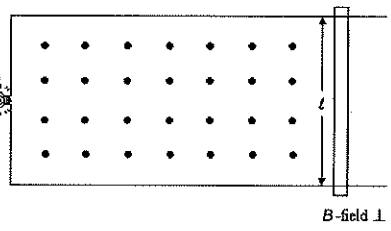
$V_{ind} = 1V$

$\phi = 90°$

$V_{ind} = BLv\sin\phi$

$v = \dfrac{V_{ind}}{BL\sin\phi} = 16,670 m/s$!

24) A conducting rod whose length is $\ell = 1.60$ m is placed on frictionless U-shaped metal rails that is connected to a lightbulb having a resistance of 4.00 Ω as shown in the figure. The rails and the rod are in the plane of the page. A constant uniform magnetic field of strength 2.20 T is applied perpendicular to and out of the paper. What is the magnitude of the external applied force needed to move the rod to the right with a constant speed of 6 m/s?



need V_{ind} to get the current in the rod.

From the current in the rod the magnetic Force on the rod due to the current can be calculated

$L = 1.6m$

$|\vec{B}| = 2.2T$

$|\vec{v}| = 6m/s$

$\phi = 90°$

$V_{ind} = BLv\sin\phi = 21.12V$

$V = IR$ $\quad I = \dfrac{V_{ind}}{R} = \dfrac{21.12V}{4\Omega} = 5.28A$

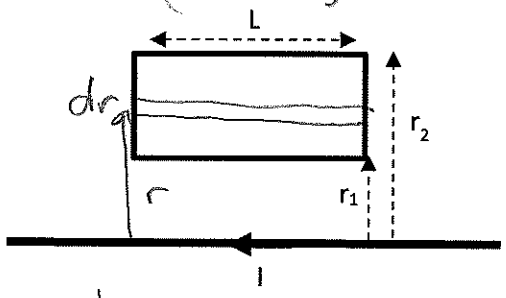
current in rod

$|\vec{F}_m| = ILB\sin\phi = (5.28A)(1.6m)(2.2T)(1)$

$|\vec{F}_m| = 18.6N$

25) A current (I) carrying long straight wire is next to a rectangular wire loop as shown. Derive the equation for the magnetic flux through the inside of the rectangular wire loop.

$$\vec{B} = \left(\frac{\mu_0 I}{2\pi r}\right)\hat{z}$$



$$\Phi_m = \int \vec{B} \cdot d\vec{A}$$

$$|\vec{B}|_{wire} = \frac{\mu_0 I}{2\pi r}$$

$$\hat{z} \cdot \hat{n} = 1$$

$$dA = L\,dr$$

$$\Phi_m = \int_{r_1}^{r_2} \frac{\mu_0 I L}{2\pi r}\,dr = \frac{\mu_0 I L}{2\pi}\int_{r_1}^{r_2} \frac{1}{r}\,dr$$

$$\Phi_m = \frac{\mu_0 I L}{2\pi}\left[\ln(x)\right]_{r_1}^{r_2} = \frac{\mu_0 I L}{2\pi}\left[\ln(r_2) - \ln(r_1)\right]$$

$$\Phi_m = \frac{\mu_0 I L}{2\pi}\ln\left(\frac{r_2}{r_1}\right)$$

KEY

1) D
2) D
3) C
4) A
5) D
6) 0.4 Wb
7) 0 Wb
8) B
9) B
10) B
11) B
12) C
13) A
14) B
15) A
16) -3.6 V
17) -57.7 V
18) 4.1 A cw
19) 12.5 V
20) 0.16 A from b to a
21) B
22) 0.6 V
23) 16,670 m/s
24) 18.6 N
25) See solutions